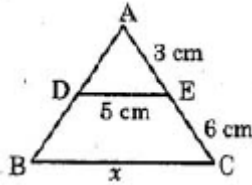


Telangana – Grade 10
2016 – Mathematics

1. In the given figure, $\triangle ABC \sim \triangle ADE$, then find the value of 'x'.



Solution:

$$\triangle ABC \sim \triangle ADE$$

$$\frac{AE}{AC} = \frac{AD}{AB} = \frac{DE}{BC}$$

$$\frac{3}{3+6} = \frac{5}{x}$$

$$x = \frac{9 \times 5}{3} = 15 \text{ cm}$$

2. Find the probability of getting a sum of the numbers on them is 7, when two dice are rolled at a time.

Solution:

$$n(s) = 36$$

$$\text{Event} = \{(1,6)(2,5)(3,4)(4,3)(5,2)(5,1)\}$$

$$p(\text{Sum}) = \frac{6}{36} = \frac{1}{6}$$

3. If $\tan \theta = \sqrt{3}$ (where θ is acute), then find the value of $1 + \cos \theta$.

Solution:

$$\tan \theta = \sqrt{3}$$

$$\tan \theta = \tan 60 \quad (\because \tan 60 = \sqrt{3})$$

$$\text{on comparing, } \theta = 60^\circ$$

$$\text{Now, } 1 + \cos \theta$$

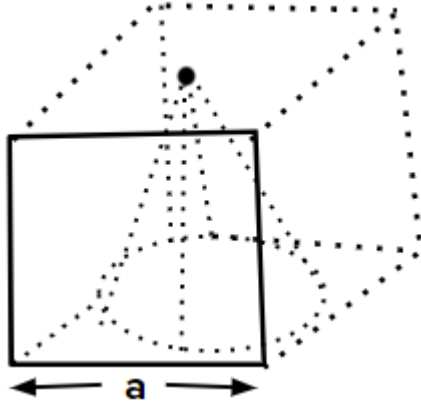
$$= 1 + \cos 60$$

$$= 1 + \frac{1}{2} = \frac{3}{2}$$

4. "A conical solid block is exactly fitted inside the cubical box of side 'a', then the volume of conical solid block is $\frac{4}{3}\pi a^3$." Is this statement true? Justify your answer.

Solution:

Diagram of cone base = a



$$\text{radius} = \frac{a}{2}$$

$$\text{Height} = a$$

$$\begin{aligned} \text{Value of cone} &= \frac{1}{3} \pi r^2 h \\ &= \frac{1}{3} \times \pi \times \left(\frac{a}{2}\right)^2 \times a \\ &= \frac{1}{12} \pi a^3 \end{aligned}$$

Hence, statement fails

5. If the surface area of hemisphere is 'S', then express 'r' in terms of 'S'.

Solution:

$$\begin{aligned} \text{Surface area of hemisphere} &= 3\pi r^2 \\ s &= 3\pi r^2 \end{aligned}$$

6. Write the formula to find the median for grouped data and explain each term.

Solution:

$$\text{Median} = \frac{l + \left(\frac{\frac{n}{2} - cf}{f}\right) \times h}{f}$$

$l \rightarrow$ lower class limit of median class

$c_f \rightarrow$ cumulative frequency preceding median class

$f \rightarrow$ frequency of median class

$h \rightarrow$ class width.

7. "If the angle of elevation of Sun increases from 0° to 90° , then the length of shadow of a tower decreases." Is this statement true? Justify your answer.

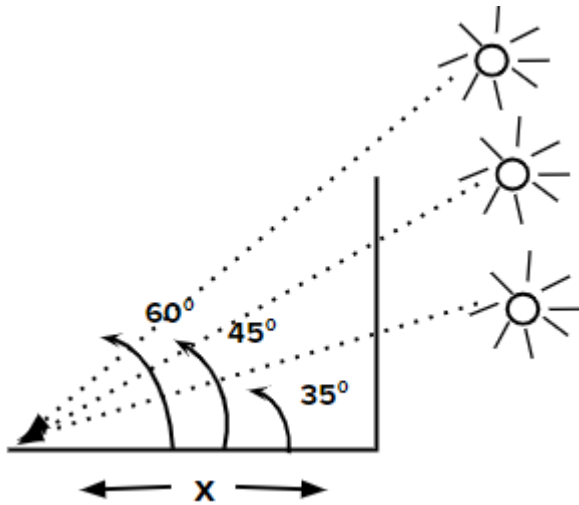
Solution:

Yes, as the angle of elevation of sun increase the length of shadow of tower decreases.

Let S take three position of sun @ 30° , 45° , 60°

Let length of shadow be x

Height of tower is H



$$@ 30^\circ \quad \tan 30^\circ = \frac{H}{x}$$

$$x = H\sqrt{3}$$

$$@ 45^\circ \quad \tan 45^\circ = \frac{H}{x}$$

$$x = H$$

$$@ 60^\circ \quad \tan 60^\circ = \frac{H}{x}$$

$$x = \frac{H}{\sqrt{3}}$$

As sun increases $@30 < @45 < @60 = H\sqrt{3} > H > \frac{H}{\sqrt{3}}$

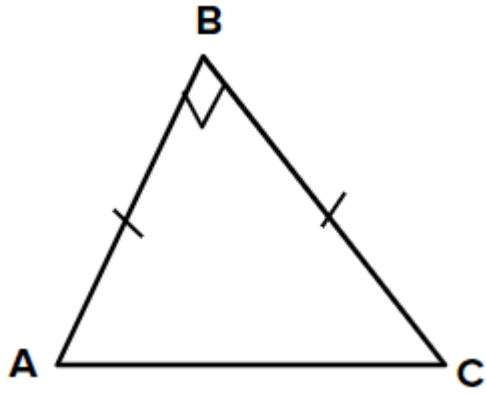
8. Prove that $\sqrt{\frac{1-\sin \theta}{1+\sin \theta}} = \sec \theta - \tan \theta$, (where θ is acute).

Solution:

$$\begin{aligned} & \sqrt{\frac{1-\sin \theta}{1+\sin \theta}} \times \frac{1-\sin \theta}{1-\sin \theta} \\ &= \sqrt{\frac{(1-\sin \theta)^2}{1-\sin^2 \theta}} = \frac{1-\sin \theta}{\cos \theta} \\ &= \sec \theta - \tan \theta \end{aligned}$$

9. ABC is an isosceles triangle and $\angle B = 90^\circ$, then show that $AC^2 = 2AB^2$.

Solution:



$$AB^2 + BC^2 = AC^2$$

$$AB^2 + AB^2 = AC^2$$

$$2AB^2 = AC^2 \quad (\because AB = BC)$$

10. Find the volume and surface area of a sphere of radius 42 cm $\left(\pi = \frac{22}{7}\right)$.

Solution:

$$\begin{aligned} \text{Volume of sphere} &= \frac{4}{3}\pi r^3 \\ &= \frac{4}{3} \times \frac{22}{7} \times 42 \times 42 \times 42 \\ &= 310464 \text{ cm}^3 \end{aligned}$$

$$\begin{aligned} \text{Surface area} &= 4\pi r^2 \\ &= 4 \times \frac{22}{7} \times 42 \times 42 \\ &= 22176 \text{ cm}^2 \end{aligned}$$

11. If $\tan(A+B)=1$ and $\cos(A-B)=\frac{\sqrt{3}}{2}$, $0^\circ < A+B < 90^\circ$ and $A > B$; find A and B.

Solution:

$$\tan(A + B) = 1$$

$$\tan(A + B) = \tan 45$$

$$A + B = 45 \quad (1)$$

$$\cos(A - B) = \frac{\sqrt{3}}{2}$$

$$\cos(A - B) = \cos 30$$

$$A - B = 30 \quad (2)$$

$$A + B = 45$$

$$\underline{A - B = 30}$$

$$2A = 75$$

$$A = 37.5^\circ$$

$$B = A - 30$$

$$= 37.5 - 30$$

$$= 7.5^\circ$$

12. A solid metallic ball of volume 64 cm^3 melted and made into a solid cube. Find the side of the solid cube.

Solution:

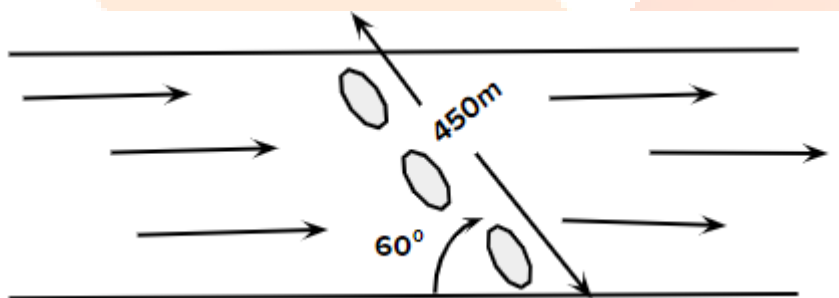
Volume of solid ball = Volume of solid cube

$$64 = a^3$$

$$a = 4$$

13. A boat has to cross a river. It crosses the river by making an angle of 60° with the bank of the river due to the stream of the river and travels a distance of 450m to reach the another side of the river. Draw the diagram for this data.

Solution:



14. A bag contains 5 red balls and some blue balls. If the probability of drawing a blue ball is double that of red ball, find the number of blue balls in the bag.

OR

Evaluate:
$$\frac{\tan^2 60^\circ + 4 \cos^2 45^\circ + 3 \sec^2 30^\circ + 5 \cos^2 90^\circ}{\csc 30^\circ + \sec 60^\circ - \cot^2 30^\circ}$$

Solution:

(a) Let number of blue balls be x

$$P(\text{Red}) = \frac{5}{5+x}$$

$$P(\text{Blue}) = \frac{x}{5+x}$$

$$\therefore \frac{x}{5+x} = 2 \times \frac{5}{5+x}$$

$$x = 10$$

(b)

$$\frac{\tan^2 60^\circ + 4 \cos^2 45^\circ + 3 \sec^2 30^\circ + 5 \cos^2 90^\circ}{\operatorname{cosec} 30^\circ + \sec 60^\circ - \cot^2 30^\circ}$$

$$= \frac{3 + 4 \times \frac{1}{2} + 3 \times \frac{4}{3} + 0}{2 + 2 - 3}$$

$$= \frac{3 + 2 + 4}{4 - 3} = 9$$

15. Consider the following distribution of daily wages of 50 workers of a factory.

Daily wages in Rupees	200-250	250-300	300-350	350-400	400-450
Number of workers	6	8	14	10	12

Find the mean daily wages of the workers in the factory by using step deviation method.

OR

Draw a circle of radius 5 cm. From a point 8 cm away from its centre, construct a pair of tangents to the circle. Find the lengths of tangents.

Solution:

(a)

Daily wages	Marks	f	d = (x - A)/i	fd
200-250	225	6	-2	-12
250-300	275	8	-1	-8
300-350	325 A	14	0	0
350-400	375	10	1	10
400-450	425	12	2	24
		50		14

$$\bar{\alpha} = A + i \left(\frac{\sum fidi}{\sum fi} \right)$$

$$= 325 + 50 \left(\frac{14}{50} \right)$$

$$= 325 + 14 = 339$$

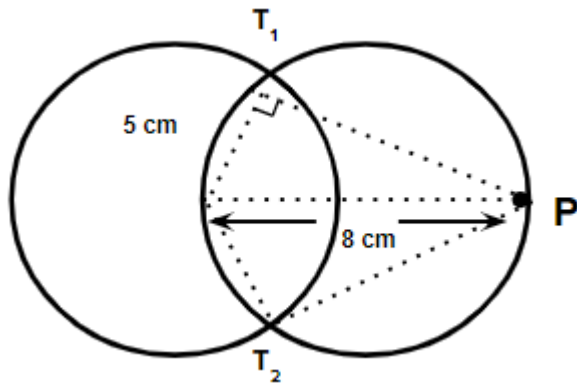
(b)

Draw circle C_1 of radius 5cm

Points P @8cm from C_1

Draw T_1P and T_2P

6.23cm



16. The following table gives production yield per hectare of wheat of 100 farms of a village.

Production yield (Quintal/Hec.)	50-55	55-60	60-65	65-70	70-75	75-80
Number of farmers	2	24	16	8	38	12

Draw both Ogives for the above data. Hence obtain the median production yield.

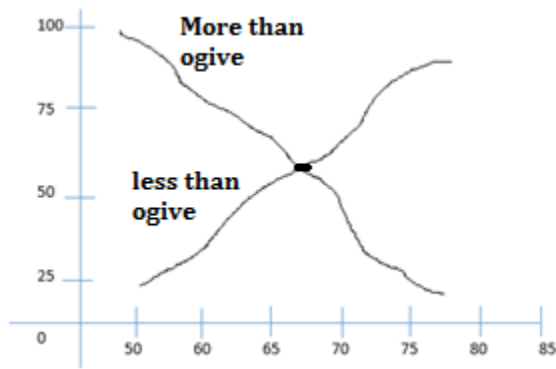
OR

Construct a triangle of sides 5 cm, 6 cm and 7cm, then construct a triangle similar to it, whose sides are $\frac{2}{3}$ of the corresponding sides of the first triangle.

Solution:

(a)

Production yield	Number of farms	Less than	Cf	More than	Cf
50-55	2	55	2	75	12
55-60	24	60	26	70	50
60-65	16	65	42	65	58
65-70	8	70	50	60	74
70-75	38	75	88	55	98
75-80	12	80	100	50	100



Median = where ever the two ogives are intersecting that point only on the graphs represents the median.

(b) Draw base = 6cm

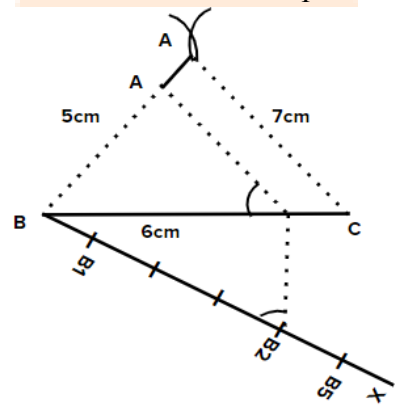
Take radius of 5 cm and drop a arc

Take radius of 7 cm and intersect a and draw triangle.

Draw line BX at an acute angle

Divide BX at equal duration to take 3 parts

Join B₂ with C' and A' parallel to base



17. DWACRA is supplied cuboidal shaped wax block with measurements $88\text{cm} \times 42\text{cm} \times 35\text{cm}$. From this how many number of cylindrical candles of 2.8 cm diameter and 8cm of height can be prepared?

OR

Two poles of equal heights are standing opposite to each other, on either side of the road, which is 80m wide. From a point between them on the road, the angles of elevation of top of the poles are 60° and 30° respectively. Find the height of the poles.

Solution:

$$\text{Volume of wax block} = 88 \times 42 \times 35 \text{ cm}^3$$

$$\text{Volume of 1 cylinder candle} = \pi \times r^2 h$$

$$= \pi \times 1.4^2 \times 8$$

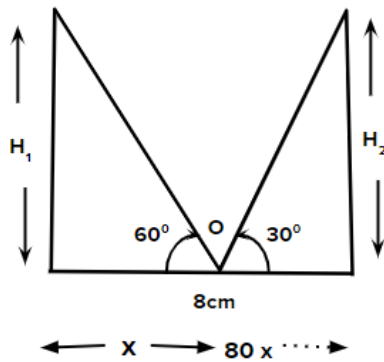
Number of candles =

$$\frac{7 \times 88 \times 42 \times 35 \times 100}{22 \times 1.4 \times 1.4 \times 8}$$

$$= \frac{15 \times 7 \times 100}{4}$$

$$= 2625$$

(b)



Let height of pole be H_1 and H_2

Let pole H_1 distance is x from 0

So, other = $80 - x$

$$\tan 60 = \frac{H_1}{x}$$

$$x = \frac{H_1}{\sqrt{3}} \quad (1)$$

$$\tan 30 = \frac{H_2}{80 - x}$$

$$80 - x = H_2 \sqrt{3}$$

$$x = 80 - H_2 \sqrt{3} \quad (2)$$

So, equate 1 & 2

$$\frac{H_1}{\sqrt{3}} = 80 - H_2 \sqrt{3} \quad (\text{since } H_1 = H_2)$$

$$H = 80\sqrt{3} - 3H$$

$$4H = 80\sqrt{3}$$

$$H = 20\sqrt{3} \text{ m}$$